

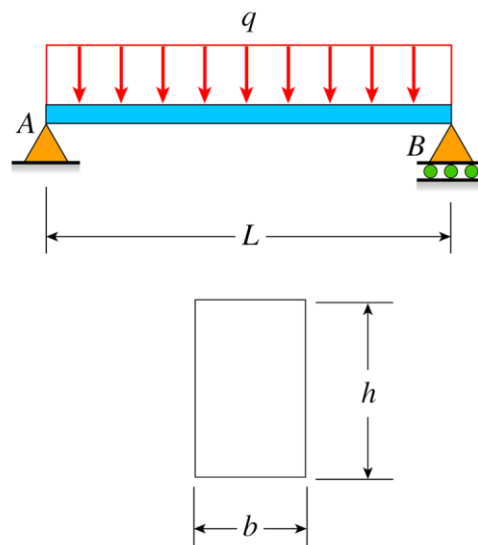
Submission: Please submit the scanned/typed document to Canvas.

1. A 1-meter-long, simply supported copper beam ($E=117$ GPa) carries uniformly distributed load q . The maximum deflection is measured as 1.5 mm.

(a) Calculate the magnitude of the distributed load q if the beam has a rectangular cross section (width $b=20$ mm, height $h=40$ mm).

(b) If instead the beam has circular cross section and $q=500$ N/m, calculate the radius r of the cross section.

Neglect the weight of the beam.



Solution:

(a) Moment of inertia

$$I = \frac{bh^3}{12} = \frac{0.02 \times 0.04^3}{12} \text{ m}^4 = 1.067 \times 10^{-7} \text{ m}^4$$

From table H.2.1, Simply supported beam $\rightarrow$ maximum deflection at midspan

$$\delta_{\max} = \frac{5qL^4}{384EI}$$

$$\Rightarrow q = \frac{384EI\delta_{\max}}{5L^4} = \frac{384 \cdot (117 \times 10^9) \cdot (1.067 \times 10^{-7}) \cdot (1.5 \times 10^{-3})}{5 \cdot (1)^4} \approx 1438.15 \text{ N/m}$$

The magnitude of the distributed load $q = 1438$ N/m

(b) Moment of inertia for circular section

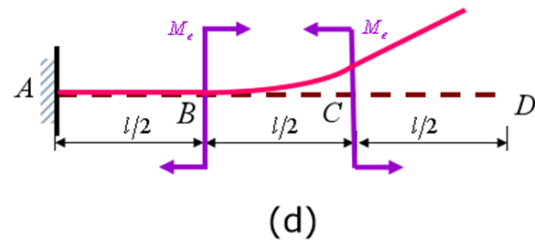
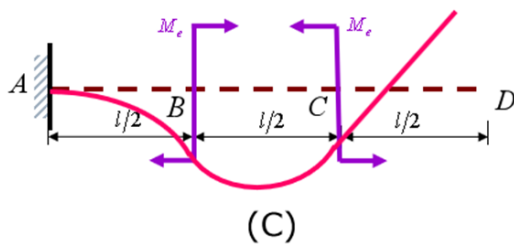
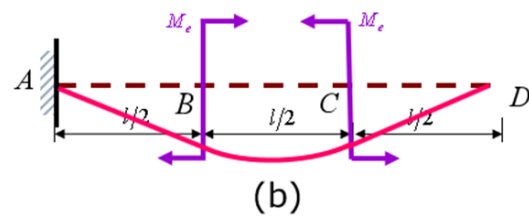
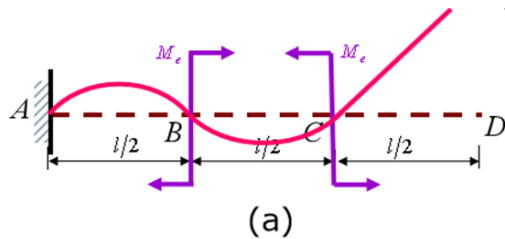
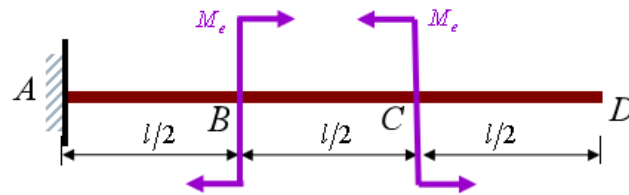
$$I = \frac{\pi r^4}{4}$$

Use same deflection formula

$$\delta_{\max} = \frac{5qL^4}{384EI} \Rightarrow I = \frac{5qL^4}{384E\delta_{\max}} \approx 3.71 \times 10^{-8} \text{ m}^4$$

$$\frac{\pi r^4}{4} = 3.71 \times 10^{-8} \text{ m}^4 \Rightarrow r \approx 14.7 \text{ mm}$$

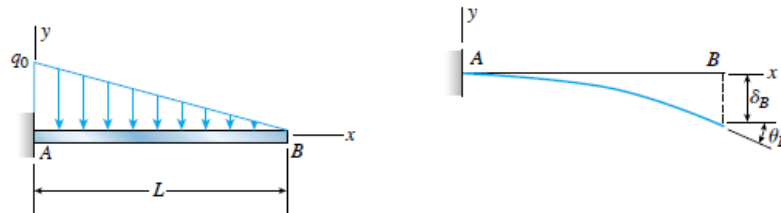
2. The cantilever beam is subjected to the loading shown in the figure. Regarding the beam's deflection curve, four possible answers are given. Please analyze and determine which one is correct.



Solution: (D)

The bending moments in segments AB and CD are zero, so these two segments remain straight without bending deformation. The deflection curves of segments AB, BC, and CD must share a common tangent at their junctions.

3. A cantilever beam AB supporting a triangularly distributed load of maximum intensity q_0 is shown in the figure. Derive the equation of the deflection curve and then obtain formulas for the deflection δ_B and deflection angle θ_B at the free end. Use the second-order differential equation of the deflection curve.



Solution: set up coordinate system like figure shown

loading intensity $q(x) = q_0(L - x)/L$

apply the load equation

$$EIw'''' = q(x) = \frac{q_0(L - x)}{L}$$

$$EIw''' = \frac{q_0}{2L}(L - x)^2 + C_1$$

shear force

$$V(x = L) = 0 \quad \text{so} \quad EI \frac{d^3w}{dx^3} \Big|_{x=L} = 0 \Rightarrow C_1 = 0$$

$$V = EI \frac{d^3w}{dx^3} = \frac{q_0}{2L}(L - x)^2$$

integration again

$$EI \frac{d^2w}{dx^2} = -\frac{q_0}{6L}(L - x)^3 + C_2$$

integrate again twice

$$EI \frac{dw}{dx} = \frac{q_0}{24L}(L - x)^4 + C_3$$

$$EIw = -\frac{q_0}{120L}(L - x)^5 + C_3x + C_4$$

$$x = 0, v' = v = 0 \Rightarrow C_3 = -\frac{q_0 L^3}{24}, \quad C_4 = \frac{q_0 L^4}{120}$$

substitute into the result.

$$\frac{dw}{dx} = -\frac{q_0 x}{24LEI}(4L^3 - 6L^2x + 4Lx^2 - x^3)$$

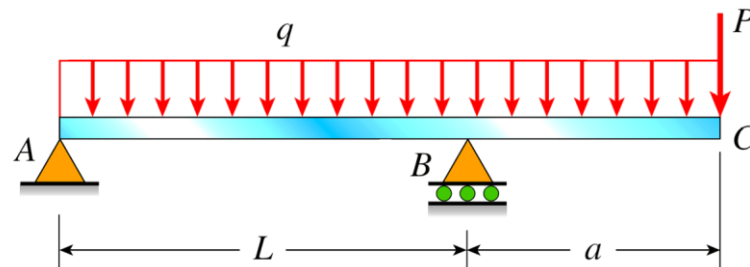
$$w = -\frac{q_0 x^2}{120LEI}(10L^3 - 10L^2x + 5Lx^2 - x^3)$$

deflection angle and deflection at the free end

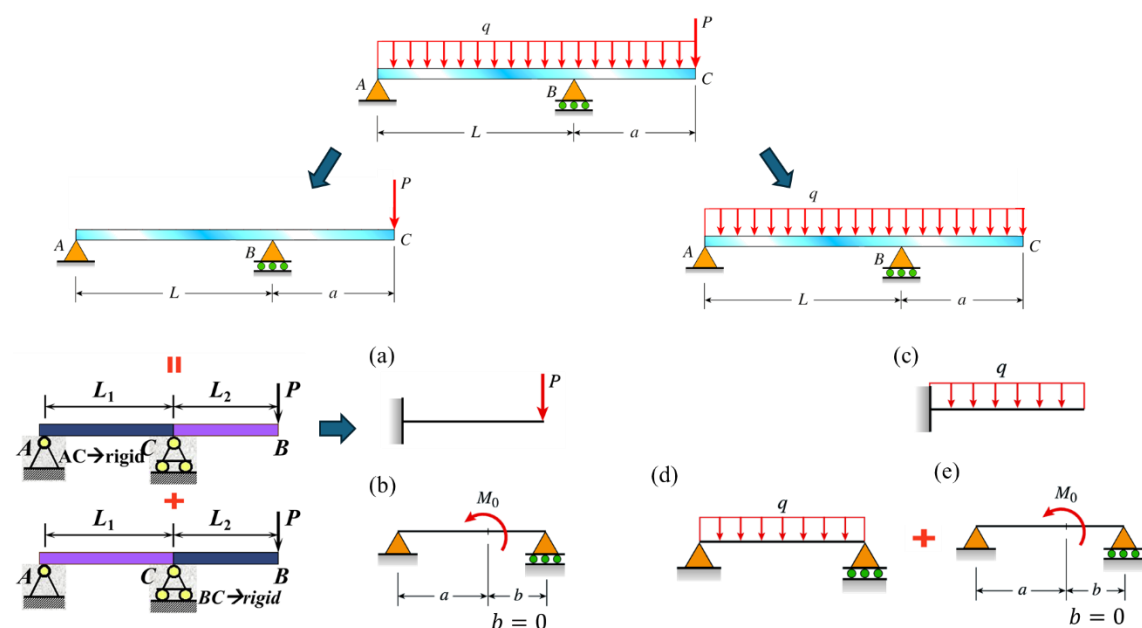
$$\theta_B = \frac{dw}{dx} \Big|_{x=L} = -\frac{q_0 L^3}{24EI}$$

$$w_B = w(x = L) = -\frac{q_0 L^4}{30EI}$$

4. Beam ABC is loaded by a uniform load q and point load P at joint C. Using the method of superposition, calculate the deflection at joint C. Assume that $L = 4$ m, $a = 2$ m, $q = 15$ kN/m, $P = 7.5$ kN, $E = 200$ GPa, and $I = 70.8 \times 10^6$ mm⁴.



Solution: Method of superposition:



(a) From H.1.4

$$v_{C1} = -\delta_{C1} = -\frac{Pa^3}{3EI} = -1.41 \text{ mm}$$

(b) From H.2.9

$$\theta_{B2} = \frac{-PaL}{3EI}$$

$$v_{C2} = a \times \theta_{B2} = \frac{-Pa^2L}{3EI} = -2.83 \text{ mm}$$

(c) From H.1.1

$$v_{C3} = -\delta_{C3} = -\frac{qa^4}{8EI} = -2.12 \text{ mm}$$

(d) From H.2.1

$$\theta_{B4} = \frac{qL^3}{24EI} = 2.825 \times 10^{-3} \text{ rad}$$

$$v_{C4} = a \times \theta_{B1} = 5.65 \text{ mm}$$

(e) From H.2.9

$$\theta_{B5} = -\frac{qa^2}{2} \frac{L}{3EI}$$

$$v_{C5} = a \times \theta_{B5} = -\frac{qa^3}{2} \frac{L}{3EI} = -5.65 \text{ mm}$$

Totally,

$$v_C = v_{C1} + v_{C2} + v_{C3} + v_{C4} + v_{C5} = -6.36 \text{ mm}$$

Solution 2: Method of successive integrations

Flexural rigidity:

$$EI = 200 \times 10^6 \text{ kN/m}^2 \times 70.8 \times 10^{-6} \text{ m}^4$$

$$= 14,160 \text{ kN.m}^2$$

Calculate the reactions:

Consider a section x-x

$$\Sigma F_y = 0$$

$$R_A + R_B - 15 \times (4 + 2) - 7.5 = 0$$

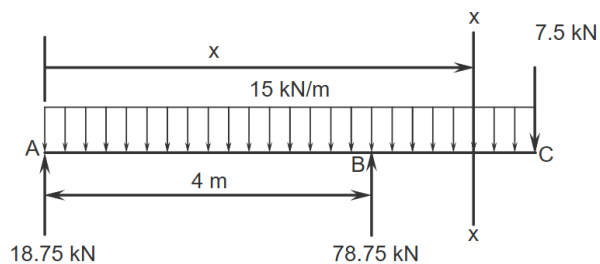
$$R_A + R_B = 97.5 \text{ kN}$$

By taking a moment about A:

$$\Sigma M_A = 0$$

$$R_B \times 4 - 15 \times 6 \times 3 - 7.5 \times 6 = 0$$

$$R_B = 78.75 \text{ kN}$$



From the equation 1,

$$78.75 + R_A = 97.5 \text{ kN}$$

$$R_A = 18.75 \text{ kN}$$

Moment about section x-x

$$M_x = 18.75x - \frac{15}{2}x^2 + 78.75 \times (x - 4)$$

From the relation,

$$EI \frac{d^2y}{dx^2} = M_x$$

$$EI \frac{d^2y}{dx^2} = 18.75x - \frac{15}{2}x^2 + 78.75 \times (x - 4)$$

Integrating twice,

$$EI \frac{dy}{dx} = \frac{18.75}{2}x^2 + C_1 - \frac{15}{6}x^3 + \frac{78.75}{2} \times (x - 4)^2$$

$$EI y = \frac{18.75}{6}x^3 + C_1x + C_2 - \frac{15}{24}x^4 + \frac{78.75}{6} \times (x - 4)^3$$

Boundary conditions:

At

$$x = 0, \quad y = 0$$

$$C_2 = 0$$

And

$$x = 4, \quad y = 0$$

$$0 = \frac{18.75}{6} \times (4)^3 + C_1 \times 4 - \frac{15}{24} \times (4)^4$$

$$C_1 = -10$$

The deflection equation:

$$EI y = \frac{18.75}{6}x^3 - 10x - \frac{15}{24}x^4 + \frac{78.75}{6} \times (x - 4)^3$$

The deflection at joint C:

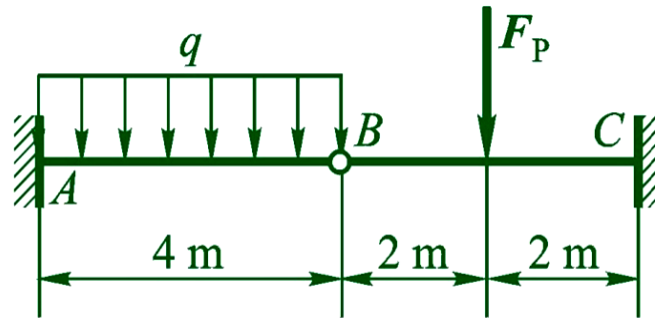
$$EI y_c = \frac{18.75}{6} \times 6^3 - 10 \times 6 - \frac{15}{24} \times 6^4 + \frac{78.75}{6} \times (6 - 4)^3$$

$$y_c = -\frac{90}{EI}$$

$$= -\frac{90 \text{ kN.m}^3}{14,160 \text{ kN.m}^2}$$

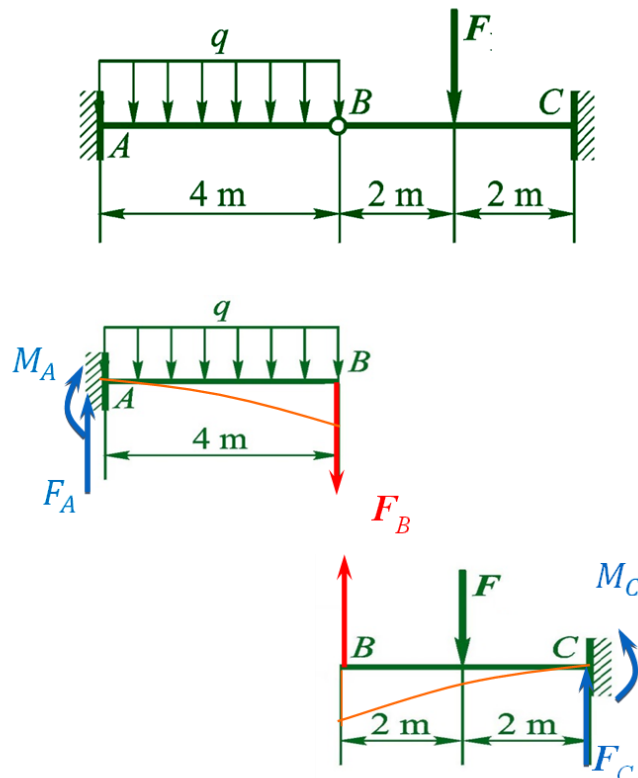
$$= -6.35 \text{ mm}$$

5. Beams AB and BC are pin-connected at B, with ends A and C fixed. Both beams have the same flexural rigidity EI . The applied concentrated force is $F=40$ kN, and the uniformly distributed load is $q=20$ kN/m. (a) calculates the vertical reaction forces and reaction bending moments on A and C ends. (b) Draw the SFD and BMD for the beam. (Hint: There are no horizontal reactions (loading is vertical only). The structure is statically indeterminate.)



Solution:

Release the structure at point B, converting the statically indeterminate structure into two cantilever beams.



Equilibrium equations:

$$F_A - F_B - 4q = 0 \quad (1)$$

$$M_A + 4q \times 2 + 4F_B = 0 \quad (2)$$

$$F_B - F + F_C = 0 \quad (3)$$

$$M_C + 2F - 4F_B = 0 \quad (4)$$

Compatibility Equation:

$$v_{B1} = v_{B2}$$

Using superposition method (H1.1, H1.4, H1.5 in deflection table)

$$v_{B1} = \frac{q \times 4^4}{8EI} + \frac{F_B \times 4^3}{3EI}$$

$$v_{B2} = \frac{F \times 2^2}{6EI} (3 \times 4 - 2) - \frac{F_B \times 4^3}{3EI}$$

Substitute into Compatibility Equation

$$\frac{q \times 4^4}{8EI} + \frac{F_B \times 4^3}{3EI} = \frac{F \times 2^2}{6EI} (3 \times 4 - 2) - \frac{F_B \times 4^3}{3EI}$$

$$\Rightarrow F_B = \frac{3}{2} \left(\frac{40 \times 10}{6 \times 4^2} - \frac{20 \times 4^4}{8 \times 4^3} \right) = -8.75 \text{ kN}$$

From equation (1), we have

$$F_A = 71.25 \text{ kN}$$

From equation (2), we have

$$M_A = -125 \text{ kN} \cdot \text{m}$$

From equation (3), we have

$$F_C = 48.75 \text{ kN}$$

From equation (2), we have

$$M_C = -115 \text{ kN} \cdot \text{m}$$

Using method of jump, the SFD and BMD:

